

Deterministic geometric Savitzky-Golay dynamics estimator

Shared parameter update: reference -> estimated

mass [kg]	2.351597591 -> 2.351597591	delta=+4.42399e-10	ratio=1
CoG x [m]	-0.002024708562 -> -0.002024708572	delta=-9.95797e-12	
ratio=-			
CoG y [m]	-3.052657894e-05 -> -3.052654922e-05	delta=+2.97162e-11	
ratio=-			
CoG z [m]	0.009509749599 -> 0.009509749619	delta=+1.98257e-11	
ratio=-			
rotor effectiveness 1	1 -> 0.9999999992	delta=-7.52697e-10	ratio=1
rotor effectiveness 2	1 -> 0.9999999994	delta=-5.82092e-10	ratio=1
rotor effectiveness 3	1 -> 0.9999999999	delta=-1.01079e-09	ratio=1
rotor effectiveness 4	1 -> 0.9999999989	delta=-1.11202e-09	ratio=1
command lag [s]	0 -> 0.004000000371	delta=+0.004	ratio=-

Inertia matrix [kg m²]

reference:

[0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[1.901508003e-05, 5.9167305e-08, 0.1289921107]

estimated:

[0.06500006149, -7.279000049e-07, 1.901508009e-05]
[-7.279000049e-07, 0.06495265636, 5.916794325e-08]
[1.901508009e-05, 5.916794325e-08, 0.1289921107]

Principal moments: reference -> estimated

principal moment 1 [kg m ²]	0.06495264516 -> 0.06495264518	delta=+1.71116e-11	
ratio=1			
principal moment 2 [kg m ²]	0.06500006701 -> 0.06500006702	delta=+9.16149e-12	
ratio=1			
principal moment 3 [kg m ²]	0.1289921163 -> 0.1289921163	delta=+8.82489e-12	ratio=1

Optimization

joint dynamics loss: 2.76279310715

Gaussian-prior cost: 2.13322787694e-16

selected strict-ZOH lag [s]: 0.004000000371

Bag failure_1_single

reconstruction position RMSE [m]: 0.000652642997664

reconstruction position component RMSE [m]: [0.00034 0.000324 0.000453]

reconstruction orientation RMSE [deg]: 0.596167087291

gyro consistency RMSE xyz [rad/s]: [0.041062 0.085837 0.026668]

specific-force consistency RMSE xyz [m/s²]: [0.92977 0.162522 0.331218]

inferred external force RMS xyz [N]: [1.982184 0.546263 4.020007]

inferred external torque RMS xyz [N m]: [0.120095 0.542534 0.119582]

bag dynamics loss: 2.76279310715

Recorded command timestamp intervals (data-derived)

Bag failure_1_single

rotor: median dt=0.00499773026 s (200.091 Hz), min=2.00271606e-05 s, max=0.0138893127 s

gimbal: median dt=0.00499868393 s (200.053 Hz), min=4.529953e-06 s, max=2.48282003 s